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1 Morse-Theoretic Atiyah-Hirzebruch Spectralsequence

1.1 Conley index

This is just a short summary. Prooves are omitted and the goal is to remaind ourself of the definitinos.

Definition 1.1.1 (Compact invariant isolated subset). For a flow $\varphi_t : M \rightarrow M$ on a locally compact metric space we call a subset $S \subseteq M$ **invariant subspace** if and only if $\varphi_t(S) = S$ for all $t \in \mathbb{R}$. For any subset $N \subseteq M$ we define the **maximal invariant subset**

$$\begin{aligned} I(N) &= \{x \in N \mid \varphi_t(x) \in N \forall t \in \mathbb{R}\} \\ &= \bigcap_{t \in \mathbb{R}} \varphi_t(N). \end{aligned}$$

A **compact invariant subset** S is called **isolated**, if a compact neighborhood N exists, such that $I(N) = S$.

Definition 1.1.2 (Index pairs). Let S be an isolated compact invariant subset. A topological pair (N, L) of compact subsets of M where $L \subseteq N$ is called an **index pair of S**, if it satisfies the following:

1. $S = I(\overline{N \setminus L}) \subseteq (N \setminus L)^\circ$.
2. $x \in L$ and $\varphi_{[0,t]}(x) \subseteq N$ implies that $\varphi_{[0,t]}(x) \subseteq L$. We call L **positively invariant in N** . Here $\varphi_{[0,t]}(x) := \{\varphi_{\tilde{t}}(x) \mid \tilde{t} \in [0, t]\}$.
3. For all $x \in N$ such that a t exists, with $\varphi_t(x) \notin N$, there exists a t' with $\varphi_{[0,t']}(x) \subseteq N$ and $\varphi_{t'}(x) \in L$.

This definition captures how the flow lines leave the invariant space S . Due to the third property we call L the **exit set**.

Definition 1.1.3 (Regular index pairs). We call an index pair (N, L) **regular** if and only if the inclusion $I : L \hookrightarrow N$ is a cofibration.

Theorem 1.1.4 (Homotopy equivalence of index pairs). *If S is an isolated compact invariant set and (N, L) and $(\tilde{N}, \tilde{L})$ are two index pairs of S . Then N/L and $\tilde{N}/\tilde{L}$ are homotopy equivalent as pointed spaces. This homotopie is canonical in transitive up to homotopy.*

Definition 1.1.5 (Connected Simple System). A **connected simple system** consists of a clection I_0 of pointed spaces along with a collection I_m of homotopy classes of maps between these spaces such that

- (i) $\text{hom}(X, Y) := \{[f] \in [X : Y] : [f] \in I_m\}$ consists of exactly one element for each pair of spaces $(X, Y) \in I_0^2$, where $[X : Y]$ denotes the set of homotopy classes of maps between X and Y .
- (ii) If $X, Y, Z \in I_0$, $[f] \in \text{hom}(X, Y)$, and $g \in \text{hom}(Y, Z)$, then $[g \circ f] \in \text{hom}(X, Z)$.
- (iii) $\text{hom}(X, Y) = \{[\text{id}_X]\}$ for all $X \in I_0$.

Remark 1.1.6. This definitions comes from Salamon in [Sal85].

Remark 1.1.7. Let S be a isolated, compact, invariant subset of M with respect to the Morse flow φ . Then the collections

$$I_0 := \{N/L : (N, L) \text{ is index pair for } S\}$$

$$I_m := \{[f] : f \text{ is the homotopy equivalence of index pairs from the proof of theorem 1.1.4}\},$$

make up a connected simple system. We call this $\mathcal{IP}_S$.

Definition 1.1.8 (Morphisms of Connected Simple Systems). A **morphism** $\Phi : I \rightarrow Y$ between two connected simple systems $I = (I_0, I_m)$ and $J = (J_0, J_m)$ is a collection of homotopy classes of maps between spaces in I_0 and J_0 such that

- (i) For every $X \in I_0$ and $Y \in J_0$ the set $\Phi(X, Y) := \{[\varphi] \in [X : Y] : [\varphi] \in \Phi\}$ consists of exactly one element.
- (ii) If $X, \bar{X} \in I_0$ and $Y, \bar{Y} \in J_0$ and if $[\varphi] \in \Phi[X, Y], [f] \in \text{hom}[\bar{X}, X], g \in \text{hom}(Y, \bar{Y})$, then $[g \circ \varphi \circ f] \in \Phi(\bar{X}, \bar{Y})$.

Remark 1.1.9. By the second property of a morphism between connected simple systems I and J we can induce such a single map $\varphi : X \rightarrow Y$ where $X \in I_0$ and $Y \in J_0$.

Remark 1.1.10 (Critical points as isolated compact invariant subsets). Let $f : M \rightarrow \mathbb{R}$ be a Morse function on a smooth Riemannian manifold (M, g) . Let $q \in \text{Crit}_k(f)$. Around q we have Morse charts $\varphi : U \rightarrow T_q M$ (after identifying q_i and ∂q_i) where $\varphi(W(q \rightarrow) \cap U) \subseteq T_q^u M$ and $\varphi(W(\rightarrow q) \cap U) \subseteq T_q^s M$. With this we can define the balls:

$$D_\varepsilon^s = \{v \in T_q^s M \mid \|v\| \leq \varepsilon\},$$

$$D_\varepsilon^u = \{v \in T_q^u M \mid \|v\| \leq \varepsilon\}.$$

They give rise to the index pair $N_q := \varphi^{-1}(D^s \times D^u)$ and $L_q = \varphi^{-1}(D^s \times \partial D^u)$. This index pair is in fact regular, since $(N_q, L_q) \cong (D^s \times D^u, D^s \times \partial D^u) \cong (D_\varepsilon^u, \partial D_\varepsilon^u)$. Now we know the K-group of such a pair:

$$K^i(N_q, L_q) = K^i(D^k, \partial D^k) = \begin{cases} \mathbb{Z} & \text{if } i - k \in 2\mathbb{N}, \\ 0 & \text{else.} \end{cases} \quad (1)$$

Notice that for $k = 0$ we need to consider $\partial D^k = \emptyset$, to get an index pair. That's why we let ∂ denote the manifold boundary insted of the topological boundary. However, now we can identify for all k :

$$C^k(M, \mathfrak{A}, \tilde{K}^l(pt)) = \bigoplus_{\text{Crit}_k(f)} \mathbb{Z} \cong \bigoplus_{q \in \text{Crit}_k(f)} K^k(N_q, L_q; \mathbb{Z}). \quad (2)$$

Or using the language of connected simple systems, we have the identification

$$C^k(M, \mathfrak{A}, \tilde{K}^l(pt)) = \bigoplus_{q \in \text{Crit}_k(f)} K^k(\mathcal{IP}_q). \quad (3)$$

define
what
K(IP)
should
mean

Definition 1.1.11 (Canonical Isomorphism Class). A **canonical isomorphism class** of rings (or other algebraic structures) is a set of rings R_0 together with a set of ring homomorphisms R_m such that

- (i) For $R, L \in R_0$ the set $\text{hom}(X, Y) := \{f : X \rightarrow Y : f \in R_m\}$ consists of exactly one element.
- (ii) If $R, L, S \in R_0$ and $f \in \text{hom}(R, L)$ and $g \in \text{hom}(L, S)$, then $g \circ f \in \text{hom}(X, Z)$.
- (iii) $\text{hom}(X, X) = \{\text{id}_x\}$

A **morphism** between two canonical isomorphism classes of rings $\Psi : (R_0, R_m) \rightarrow (S_0, S_m)$ is a collection of homomorphisms such that

1. For any $R \in R_0$ and $S \in S_0$ the set $\Psi(R, S) := \{f : R \rightarrow S : f \in \Psi\}$ contains exactly one element.
2. If $R, \bar{R} \in R_0$ and $S, \bar{S} \in S_0$, $f \in \text{hom}(\bar{R}, R)$, $g \in \text{hom}(S, \bar{S})$ and $\psi \in \Psi(R, S)$, then $g \circ \psi \circ f \in \Psi(\bar{R}, \bar{S})$.

Remark 1.1.12. Again the second property lets us induce a morphism between two canonical isomorphism classes of rings $\Psi : (R_0, R_m) \rightarrow (S_0, S_m)$ by a single isomorphism

$$\psi : R \rightarrow S$$

for $R \in R_0$ and $S \in S_0$.

Remark 1.1.13. We call the category of simple connected systems **SCS** and the categorie of canonical isomorphism classes of rings **CICR**

Definition 1.1.14. The classical complex topological K-functors $\tilde{K}^i$ define a family of kontravariant functors

$$\text{SCS} \rightarrow \text{CICR}$$

that is fiven on the objects as follows: For (I_0, I_m) being a family of pointed topological spaces, we get a new family of rings as:

$$R_0 := \{\tilde{K}^i(X) : X \in I_0\},$$

and a family of ringhomomorphisms

$$R_m := \{\tilde{K}^i(f) : [f] \in I_m\}.$$

Obviously, the tuple (R_0, R_m) is an object in **CICR**. Now a morphism in $\Phi : (I_0, I_m) \rightarrow (J_0, J_m)$ in **SCS** gives rise to a collecting of ringisomorphisms

$$\Psi := \{\tilde{K}^i(\varphi) : [\varphi] \in \Phi\}.$$

By the homotopy property of reduced K-theory, this becomes a morphism in **CICR**.

Example 1.1.15. Any pointed topological space X defines a simple connected system $(\{X\}, [\text{id}_x])$. Similar, any ring R defines a canonical isomorphism class of rings $(\{R\}, \{\text{id}_R\})$.

Definition 1.1.16. Let (M, f, g) be a smooth manifold M together with a Morse Smale function f . Let $S \subseteq M$ be a compact invariant subset. Then the index pairs of S together with the canonical isomorphisms between them define a simple connected system which we will denote with $\mathcal{IP}_S$.

Remark 1.1.17. Let $\underline{\mathbb{Z}}$ be the canonical equivalence class of rings consisting only of the ring $\mathbb{Z}$, and let p be a critical point of index λ_p in M . Then remark 1.1.10 gives us the existence of a morphism for all $i \in \mathbb{N}$:

$$\tilde{K}^{\lambda_p+2i}(\mathcal{IP}_p) \rightarrow \underline{\mathbb{Z}}.$$

The next remark will help us to get a canonical isomorphism.

Remark 1.1.18. Furthermore, this isomorphism

$$C^k(M, \mathfrak{A}, \tilde{K}^l(pt)) = \bigoplus_{\text{Crit}_k(f)} \mathbb{Z} \cong \bigoplus_{q \in \text{Crit}_k(f)} K^k(N_q, L_q; \mathbb{Z}). \quad (4)$$

can be made canonically, using orientations: Let q be a critical point in index λ_q . Let $\mathfrak{b}$ be a ordered basis of $T_q^u M$. Let ψ be a refined Morse chart around q , such that the induced basis in $T_q M = T_q^u M \oplus T_q^s M$ projects onto a basis of the first factor $T_q^u M$ that is compatible with $\mathfrak{b}$. Using this Morse chart do define the generator of the regular index pair of q gives a unique generator.

Proof. Let ψ_1 and ψ_2 be two such Morse charts. Notice that they only differ by order of the directions and their signs, meaning that for all i there is a j such that $\psi_1^{-1}(e_i) = \pm \psi_2^{-1}(e_j)$. Hence, viewed as maps into the tangent space we have that $\psi_1(N_p, L_q) = \psi_2(N_p, L_q) \subseteq T_q M$. We define $B_\varepsilon^{\lambda_q}(0) \subseteq \mathbb{R}^{\lambda_q}$ to be the sphere. The generator of $\psi_1(N_p, L_q)$ comes as the pullback of the canonical Bott generator of $K^i(B_\varepsilon^{\lambda_q}(0), \partial B_\varepsilon^{\lambda_q}(0))$ via the basis map after the collapse map $c : (D^s \times D^u, D^s \times \partial D^u) \cong (D_\varepsilon^u, \partial D_\varepsilon^u)$. Let α be a generator of $\psi_1(N_p, L_q)$ induced from the basis $\mathfrak{b}_1$ that comes from ψ_1 . I.e.

$$\alpha = (B_{\mathfrak{b}_1} \circ c)^*(\beta)$$

By assumption, the map $B_{\mathbf{b}_1} \circ B_{\mathbf{b}}^{-1}$ has determinant one. Hence it is homotopic to the identity, and this homotopy can be reduced to a well defined homotopy on the quotient space $(D_\varepsilon^u / \partial D_\varepsilon^u)$. Hence

$$\alpha = (B_{\mathbf{b}} \circ c)^* (\beta)$$

And thereby, the generator only depends on $\mathbf{b}$ concluding the proof. $\square$

Example 1.1.19 (A Map of K-Groups). Let $f : M \rightarrow \mathbb{R}$ be a Morse Smale function on a smooth compact Riemannian manifold (M, g) . Let $q \in \text{Crit}_k(f)$ and $p \in \text{Crit}_{k-1}(f)$. Assume for a moment we already have an isolated regular compact index pair (N_2, N_0) of $S = W(p \rightarrow q) \cup \{p, q\}$. For a $c \in (f(p), f(q))$ we set $N_1 = N_0 \cup (N_2 \cap M^c)$, where M^c is the sublevel set. Clearly now (N_2, N_1) is an index pair for q and (N_1, N_0) is a pair for p . Assuming we have regular index pairs (N_q, L_q) and (N_p, L_p) for p, q we can now define a map:

$$\Delta^k(q \rightarrow p) : K^{k-1}(N_p, L_p) \rightarrow K^k(N_q, L_q)$$

as the composition:

$$K^{k-1}(N_p, L_p; \mathbb{Z}) \xrightarrow{\cong} K^{k-1}(N_1, N_0; \mathbb{Z}) \xrightarrow{\delta_{\text{triple}}} K^k(N_2, N_1; \mathbb{Z}) \xrightarrow{\cong} K^k(N_q, L_q; \mathbb{Z})$$

where δ_{triple} is the connecting homomorphism, and the other two maps are induced by the homotopic equivalence of two index pairs corresponding to the same isolated compact invariant subset. Putting all together we can define a morphism:

$$\Delta^k : \bigoplus_{p \in \text{Crit}_{k-1}(f)} K^{k-1}(N_p, L_p) \rightarrow \bigoplus_{q \in \text{Crit}_k(f)} K^k(N_q, L_q) \quad (5)$$

We can define this also with canonical isomorphism classes:

$$\underline{\Delta}^k : \bigoplus_{p \in \text{Crit}_{k-1}(f)} K^{k-1}(\mathcal{IP}_p) \rightarrow \bigoplus_{q \in \text{Crit}_k(f)} K^k(\mathcal{IP}_q)$$

Definition 1.1.20 (Regular Index Pairs and a Filtration on M). As always let $f : M \rightarrow \mathbb{R}$ be a Morse-Smale function on a compact smooth Riemannian manifold (M, g) of dimension m . Let $\varphi_t : M \rightarrow M$ be the flow given by $-\text{grad} f$. For $0 \leq j \leq k \leq m$ define:

$$W(k, j) = \bigcup_{j \leq \lambda_p \leq \lambda_q \leq k} W(q \rightarrow p).$$

There exists a filtration

$$\emptyset =: N_{-1} \subset N_0 \subset N_1 \subset \dots \subset N_{m-1} \subset N_m = M, \quad (6)$$

such that (N_k, N_{j-1}) is a regular index pair for $W(k, j)$ for all $0 \leq j \leq k \leq m$. If we define $\mathcal{IP}_i := \mathcal{IP}_{\text{Crit}_i(f)}$ we can induce a filtration of connected simple systems:

$$\emptyset \rightarrow \mathcal{IP}_0 \rightarrow \mathcal{IP}_1 \rightarrow \dots \rightarrow \mathcal{IP}_{m-1} \rightarrow \mathcal{IP}_m$$

define directly sum of canonical isomorphism classes

Proof. The proof is done by Dietmar Salamon and can be checked in [Sal85] corollary 4.4. $\square$

Remark 1.1.21. Now $\mathcal{IP}_k$ has a pointed spaces as a representative, that if of the form $\bigvee_{p \in \text{Crit}_k} (N_p/L_p)$. Hence, $\tilde{K}^k(\mathcal{IP}_k)$ has a representative that omitts an isomorphism to $\bigoplus_{p \in \text{Crit}_k(f)} \tilde{K}^k(N_p/L_p)$. Hence as canonical isomophism classes of rings we get the isomorphism:

$$\tilde{K}^k(\mathcal{IP}_k) \rightarrow \bigoplus_{q \in \text{Crit}_k(f)} \tilde{K}^k(\mathcal{IP}_p)$$

and with this we can define the morphism $\underline{\Delta}^k : \tilde{K}^{k-1}(\mathcal{IP}_{k-1}) \rightarrow \tilde{K}^k(\mathcal{IP}_k)$. I think with this, i can give the definition of the following spectral sequence in the language of canonical isomorphism classes!

2 The Spectral Sequence using Exact Couples

Definition 2.0.1. Let M be a smooth manifold with Morse data and

$$\emptyset =: N_{-1} \subset N_0 \subset N_1 \subset \dots \subset N_{m-1} \subset N_m = M,$$

be a filtrations constructed in 1.1.20. Then we define the Groups

$$\begin{aligned} D^{p,q} &:= K^{p+q}(N_p) & \text{and} & \quad D := \bigoplus_{p,q} D^{p,q} \\ E^{p,q} &:= K^{p+q}(N_p, N_{p-1}) & \text{and} & \quad E := \bigoplus_{p,q} E^{p,q} \end{aligned}$$

Now we have the exact sequence from the pair (N_p, N_{p-1}) :

$$\dots \longrightarrow K^{p+q-1}(N_{p-1}) \xrightarrow{\delta_{p,q}^*} K^{p+q}(N_p, N_{p-1}) \xrightarrow{j_{p,q}^*} K^{p+q}(N_p) \xrightarrow{i_{p,q}^*} K^{p+q}(N_{p-1}) \dots$$

With this we define the maps

$$\begin{array}{lcl} \alpha & := \bigoplus_{p,q} i_{p,q}^* : D \rightarrow D & (-1, 1) \\ \beta & := \bigoplus_{p,q} \delta_{p,q}^* : D \rightarrow E & (1, 0) \\ \gamma & := \bigoplus_{p,q} j_{p,q}^* : E \rightarrow D & (0, 0) \end{array} \quad \left| \begin{array}{l} \\ \\ \text{bidegree} \end{array} \right.$$

Obviously by definition the compositions in the triangle are exact:

$$\begin{array}{ccc} D & \xrightarrow{\alpha} & D \\ & \swarrow \gamma & \searrow \beta \\ & E & \end{array}$$

Finally we define $d := \beta \circ \gamma$.

Definition 2.0.2 (Exact Couples). Two groups D, E and maps

$$\begin{array}{lcl} \alpha: & D \rightarrow D & D \xrightarrow{\alpha} D \\ \beta: & D \rightarrow E & \swarrow \gamma \quad \searrow \beta \\ \gamma: & E \rightarrow D & E \end{array}$$

such that the compositions in the triangle are exact are called an **exact couple**. we define $d := \beta \circ \gamma$. In this case it is obvious that $d^2 = 0$.

Definition 2.0.3 (Categorie of Exact Couples). We define the categorie **ExCoup** as follows: The objects are exact couples, i.e. quintibles $(D, E, \alpha, \beta, \gamma)$ where D, E are groups (or objects in an abelian categorie), such that the triangle is exact:

$$\begin{array}{ccc} D & \xrightarrow{\alpha} & D \\ & \swarrow \gamma \quad \searrow \beta & \\ & E & \end{array}$$

.Given another exact couple $(D', E', \alpha', \beta', \gamma')$ we define a morphism from one to another to be a tuple of morphisms $(f : D \rightarrow D', g : E \rightarrow E')$ such that the following diagram commutes:

$$\begin{array}{ccccccc} D & \xrightarrow{\alpha} & D & \xrightarrow{\beta} & E & \xrightarrow{\gamma} & D \\ \downarrow f & & \downarrow f & & \downarrow g & & \downarrow f \\ D' & \xrightarrow{\alpha'} & D' & \xrightarrow{\beta'} & E' & \xrightarrow{\gamma'} & D' \end{array}$$

Corollary 2.0.4 (An Endofunktor in the Categorie of Exact Couples). We can derive a new exact couple with the following definitions:

$$\begin{array}{lll} E' & := \ker d / \text{imd} & \alpha' : D' \rightarrow D' \quad a \mapsto \alpha(a) \\ D' & := \alpha(A) & \beta' : D' \rightarrow E' \quad a = \alpha(\tilde{a}) \mapsto [\beta(\tilde{a})] \\ & & \gamma' : E' \rightarrow D' \quad [b] \mapsto \gamma(b) \end{array} \quad \begin{array}{ccc} D' & \xrightarrow{\alpha'} & D' \\ & \swarrow \gamma' \quad \searrow \beta' & \\ & E' & \end{array}$$

We again define $d' := \beta' \circ \gamma'$. The derifing of a new exact couple as defined above defines an endofunktor which er denote by Der .

Proof. Let $(D, E, \alpha, \beta, \gamma)$ be the exact couple we started with and call the derived couple $(D', E', \alpha', \beta', \gamma')$. First we show the well definition of β . Let $a \in D' = \alpha(a_1) = \alpha(a_2)$. Then $a_1 - a_2 \in \ker \alpha = \text{imd}$. Hence $\beta(a_1) - \beta(a_2) = \beta(a_1 - a_2) \in \text{im}(\beta \circ \gamma) = \text{imd}$. Furthermore, since $\beta(a_1) \in \text{im} \beta = \ker \gamma \subseteq \ker d$ we have that the map β' is well defined. For the well definition of γ' we ssume that $[b_1] = [b_2] \in E'$. But then the difference of b_1 and b_2 is zero in E' meaning in the image oif d , which is again contained in the image of β and thereby in the kernal of α . This menas that under α the image of b_1 and b_2 agree. Now to show that the derived couple is also exact one continuous the diagram yoga. We do exemplary the equality of the kernal of β' with the image of α' : Assume

that $\beta'(x) = 0$. That means that $\beta(\tilde{x}) \in \text{imd}$ with $\alpha(\tilde{x}) = x$. We now want to show that $x \in \alpha^2 D$, and do this by showing that $\tilde{x} = \alpha(z)$ for some z . Now since $\beta(\tilde{x}) \in \text{imd}$ we have that $\beta(\tilde{x}) = d(y) = \beta \circ \gamma(x)$ for some y . In other words, $\tilde{x} - \gamma(y) \in \ker \beta = \text{im} \alpha$. Hence, $\tilde{x} - \gamma(y) = \alpha(z)$ for some z . But now applying α to both side and using that $\alpha \circ \gamma = 0$ we can deduce that $\alpha^2(z) = \alpha(\tilde{x})$. This shows that the kernel of β' is contained in the image of α' . The other direction of the inclusion is easier: Let $a = \alpha(\tilde{a}) \in D'$. Then $\beta' \circ \alpha'(z) = [\beta(z)] = \beta \circ \alpha(\tilde{a}) = 0$. Similarly one checks for exactness in the other corners. Now to show that we have an endofunctor we need to define what to do on morphisms. So assume that we have the two exact couples $(D, E, \alpha, \beta, \gamma)$ and $(\tilde{D}, \tilde{E}, \tilde{\alpha}, \tilde{\beta}, \tilde{\gamma})$ with the morphisms $(f : D \rightarrow \tilde{D}, g : E \rightarrow \tilde{E})$. We now need to define a morphism of exact couples $((f' : D' \rightarrow \tilde{D}', g' : E' \rightarrow \tilde{E}'))$ between the derived couples. For this we define:

$$f' : D' \rightarrow \tilde{D}'; x \mapsto f(x) \quad \text{and} \quad g' : E' \rightarrow \tilde{E}'; [x] \mapsto [g(x)]$$

and check for well definition as maps and that the pair is a morphism of exact couples. The well definition of f' follows directly from $f \circ \alpha = \tilde{\alpha} \circ f$. For the well definition of g' assume that $[x] = [y] \in E'$, meaning that $x - y \in \text{imd} = \text{im} \beta \circ \gamma$ hence for some z we have that $x - y = \beta \circ \gamma(z)$ and hence $g(x - z) = g \circ \beta \circ \gamma(z) = \tilde{\beta} \circ f \circ \gamma(z) = \tilde{\beta} \circ \tilde{\gamma} \circ g(z) \in \text{imd}$. This proves the well definition. $\square$

Definition 2.0.5 (Spectral Sequence From an Exact Couple). We can now iterate this procedure to get a list of spaces with maps $(E, d), (E', d'), (E'', d'') \dots$ or often written as $(E_1, d_1), (E_2, d_2), \dots$ which is called a **spectral sequence**. Notice how E_{r+1} is given by (E_r, d_r) whilst d_{r+1} needs further information. Whilst the spectral sequence is defined inductively, the needed information is not captured in $(E_r, d_r)_r$ but rather in the exact couple. Or said differently, we get a sequence of exact couples $(E_r, D_r, \alpha_r, \beta_r, \gamma_r)$ from which we can get the spectral sequence. We can reformulate the maps as follows:

$$D_r = \alpha^{r-1}(D), \quad E_r = \ker d_{r-1} / \text{imd}_{r-1}, \quad \alpha_r = \alpha|_{\alpha^{r-1}(D)}, \quad \beta_r = [\beta \circ \alpha^{1-r}], \quad \gamma_r = \gamma, \quad d_r = \beta_r \circ \gamma_r$$

α^{1-r} means taking the preimage by alpha $1-r$ times. The well definition of this formulation was proven earlier. The bidegree of the maps is easy to deduce from the formulations above:

$$\begin{array}{ll} \alpha_r & \text{has bidegree} \quad (-1, 1) \\ \beta_r & \text{has bidegree} \quad (-r + 2, r - 1) \\ \gamma_r & \text{has bidegree} \quad (0, 0) \\ d_r & \text{has bidegree} \quad (r, r + 1) \end{array}$$

Corollary 2.0.6. We can define the category of spectral sequences **SpS** where the objects consist of families of pairs $(E_r, d_r)_{r \in \mathbb{Z}}$ where E_r are groups (or objects in an abelian category) and d_r are endomorphisms in E_r , such that $E_{r+1} \cong \frac{\ker d_r}{\text{imd}_r}$. Morphisms between two spectral sequences $(E_r, d_r)_{r \in \mathbb{N}} \rightarrow (\tilde{E}_r, \tilde{d}_r)_{r \in \mathbb{N}}$ are families of morphisms $(f_r : E_r \rightarrow \tilde{E}_r)$ such that for all r we have that $f_r \circ d_r = \tilde{d}_r \circ f_r$.

Corollary 2.0.7. We have a functor from **ExCoup** to **SpS** given as follows: Let $(E, D, \alpha, \beta, \gamma)$ be an exact couple. Let $(E_r, d_r, \alpha_r, \beta_r, \gamma_r) = \text{Der}^r((E, D, \alpha, \beta, \gamma))$ where

Der^r denotes the r -th derivation of the exact couple. Then $(E_r, d_r := \beta_r \circ \gamma_r)_r$ is an exact sequence as described above. Assume that we have the two exact couples $(D, E, \alpha, \beta, \gamma)$ and $(\tilde{D}, \tilde{E}, \tilde{\alpha}, \tilde{\beta}, \tilde{\gamma})$ with the morphisms $(f : D \rightarrow \tilde{D}, g : E \rightarrow \tilde{E})$. Then the family $(Der^r(f))_r$ is a morphism of spectral sequences.

Corollary 2.0.8 (Explicit Pages). For a spectral sequence arising from an exact couple defined as above we have a natural isomorphism

$$E_{r+1} = \frac{\gamma^{-1}(\alpha^r D)}{\beta(\ker \alpha^r)}$$

Proof. We proceed by induction starting with $n = 1$. Here we have by exactness:

$$\ker(\beta \circ \gamma) = \gamma^{-1}(\ker(\beta)) = \gamma^{-1}(\operatorname{im}(\alpha)), \quad \text{and} \quad \operatorname{im}(\beta \circ \gamma) = \beta(\ker(\alpha)).$$

Hence:

$$E_2 = \frac{\ker(d_1)}{\operatorname{im}(d_1)} = \frac{\ker(\beta \circ \gamma)}{\operatorname{im}(\beta \circ \gamma)} = \frac{\gamma^{-1}(\alpha D)}{\beta(\ker(\alpha))}.$$

Now Assume that

$$E_r = \frac{\gamma^{-1}(\alpha^{r-1} D)}{\beta(\ker \alpha^{r-1})}$$

Then we have

$$\ker d_r = \ker \beta_r \gamma_r = \gamma_r^{-1}(\operatorname{im}(\alpha_r)) = \gamma_r^{-1}(\alpha^r(D)).$$

Now $\gamma_r : E_r \rightarrow D_r$ maps $[x] \mapsto \gamma(x)$, hence by induction assumption

$$\ker d_r = \frac{\gamma^{-1}(\alpha^r D)}{\beta(\ker \alpha^r)}.$$

Notice that $\alpha^{r-1}(\ker(\alpha_r))$ contains those elements in D , that are killed by the α^n , in other words this equals $\ker \alpha^r$. Now $\beta_r = [\beta \circ \alpha^{r-1}]$ and hence using the induction assumption again we can conclude:

$$\operatorname{im}(\beta_r \gamma_r) = [\beta(\alpha^{r-1} \ker(\alpha_r))] = \frac{\beta(\ker \alpha^r D)}{\beta(\ker \alpha^{r-1})}.$$

Now using the isomorphism theorem we get the natural isomorphism from the statement. $\square$

Definition 2.0.9. In our case all pages of the spectral sequence inherit a double grading and the boundary maps to. Hence we will denote

$$E_r = \bigoplus_{p,q} E_r^{p,q} \quad \text{and} \quad d_r = \oplus_{p,q} d_r^{p,q}$$

Corollary 2.0.10. The grading carries through all pages and we can reformulate

$$E_{r+1}^{p,q} = \frac{\gamma^{-1} \alpha^r D}{\beta(\ker \alpha^r)} \cap E_{r+1}^{p,q} = \frac{\gamma^{-1}((\alpha^r D) \cap D^{p,q})}{\beta((\ker \alpha^r) \cap D^{p-1,q})} = \frac{\gamma^{-1} \alpha^r D^{p+r, q-r}}{\beta(\ker \alpha^r|_{D^{p-1,q}})}$$

Corollary 2.0.11 (The Exact Sequence Stabilizes). For all $r \geq m + 1$ it holds that

$$E_r = E_{r+1} = \dots$$

For this great enough r we define $E_\infty := E_r$.

Proof. Notice how our original $E^{p,q}$ is supportet in a strip regarding p which is fancy language for saying that $E^{p,q} = 0$ for $p > 0$ and for $p \geq m$. These conditinos cary through the whole spectral sequence, as succeeding pages are quotients of preceeding ones respecting the grading. Now the map d_r has bidegree $(r, -r + 1)$, and hence $d_r = 0$ since the domain or codomain is zero, if the p grading is greater then the p -length of the supporting strip meaning $p > m$ which gives us the claimes $r \geq m + 1$. $\square$

Now we refomulate the corollary 2.0.8 for the infinity page. We have the constant sequence of groups $E_r = E_{r+1} = \dots$ and by the corollary we have

$$\frac{\gamma^{-1}(\alpha^r D)}{\beta(\ker \alpha^r)} = \frac{\gamma^{-1}(\alpha^{r+1} D)}{\beta(\ker \alpha^{r+1})} = \frac{\gamma^{-1}(\alpha^{r+2} D)}{\beta(\ker \alpha^{r+2})} = \dots$$

Hence in our later inspections of E_∞ we can assume r to be arbitrary big.

Lemma 2.0.12 (The last page). *In our spectral sequence, we have that*

$$E_\infty^{p,q} = \frac{\ker[K^{p+q}(M) \rightarrow K^{p+q}(N_{p-1})]}{\ker[K^{p+q}(M) \rightarrow K^{p+q}(N_{p1})]}$$

Proof. For each fixed (p, q) there is an r big enough such that $\alpha^r D^{p,q} = 0$, since α^r has degree $-r$ in p , hence for $r > p$, α^r maps $D^{p,q}$ to something negative in p and hence to the trivial group. Thus $D = \cup_{r \geq 0} \ker \alpha^r$. Now the denominator of $E_r^{p,q}$ equals $\beta(\ker(\alpha^r|_{D^{p+q-1,q-r}}))$ and hence for r big enough this equals the image of $\beta|_{D^{p-1,q}}$ which in turn is the kernal of $\gamma|_{E^{p,q}}$. For the numerator, we start by looking at $(\alpha^r D)^{p,q} = \alpha^r D^{p+r,q-r}$. Now $D^{p+r,q-r} = K^{p+q}(N_{p+n})$. Hence for n big enough (such that $N_{p+n} = M$) we can calculate $\alpha^n(K^{p+q}(N_{p+n})) = \text{im}(K^{p+q}(M) \rightarrow K^{p+q}(N_p))$. This results in the description

$$E_\infty^{p,q} = \frac{\gamma^{-1}(\text{im}(K^{p+q}(M) \rightarrow K^{p+q}(N_p)))}{\ker \gamma|_{E^{p,q}}}$$

Next we define the groups

$$L^{p,q} := \text{im}(K^{p+q}(M) \rightarrow K^{p+q}(N_p)) \cap \text{im} \gamma = \text{im}(K^{p+q}(M) \rightarrow K^{p+q}(N_p)) \cap \ker \alpha$$

And since we have for any f and any subset S of the codomain of f the equality $f^{-1}(S) = f^{-1}(S \cap \text{im} f)$ holds, we have in our setting $E_\infty^{p,q} = \frac{L^{p,q}}{\ker(\gamma|_{E^{p,q}})}$. Now the kernal of the funktion

$$\tilde{\gamma} : \gamma^{-1}(L^{p,q}) \rightarrow L^{p,q}; x \mapsto \gamma(x)$$

is das wahr mit gamma eingeschränkt auf $E^{p,q}$?

equals $\ker \gamma|_{E^{p,q}}$: Since $\gamma^{-1}(L^{p,q}) \subseteq E^{p,q}$ the kernel of $\tilde{\gamma}$ is clearly contained in $\ker \gamma|_{E^{p,q}}$. If on the other hand $\gamma(x) = 0$ for $x \in E^{p,q}$ then $x \in \gamma^{-1}(0)$ and since zero is contained in $L^{p,q}$ we conclude that the two kernels are equal and hence we can apply the isomorphism theorem to get the isomorphism that is induced from γ :

$$E_{\infty}^{p,q} = \frac{L^{p,q}}{\ker(\gamma|_{E^{p,q}})} \rightarrow L^{p,q}.$$

Now finally we define the mapping

$$\ker(K^{p+q}(M) \rightarrow K^{p+q}(N_{p-1})) \rightarrow L^{p,q} = \text{im}(K^{p+q}(M) \rightarrow K^{p+q}(N_p)) \cap \ker \alpha$$

That maps an element $x \in K^{p,q}(M)$ to the pullback of the inclusion $N_p \rightarrow M$. This map is well defined, since applying α to the image of an element from the domain is the same as the pullback of x via the inclusion $N_{p-1} \rightarrow M$ and equals zero by definition of the domain. Hence the original map maps into the kernel of alpha which proofs the well definition. This line of reasoning also shows the surjectivity since every element in $K^{p,q}(N_p)$ that comes from the whole M and maps to zero if restricted to N_{p-1} can be expressed as the pullback from the whole M restricted to those, that vanish over N_{p-1} . Its easy to see that the kernel of this surjection equals the kernel of the pullback

$$K^{p,q}(M) \rightarrow K^{p,q}(N_p),$$

because ever element that pulls back to zero over N_p also pulls back to zero over N_{p-1} , and the other containemend is trivially true. To the surjection we can again apply the isomorphism theorem to get the isomorphism

$$\frac{\ker(K^{p+q}(M) \rightarrow K^{p+q}(N_{p-1}))}{\ker(K^{p,q}(M) \rightarrow K^{p,q}(N_p))} \cong L^{p,q},$$

that is induced by the pullback of the inclusion. Together with the γ induced isomorphism from $E_{\infty}^{p,q} \rightarrow L^{p,q}$ we conclude the proof. $\square$

Definition 2.0.13 (Associated Graded Group). Let G be a group together with a finite filtration of subgroups $\{F^i\}_{i=-1, \dots, -m}$ for $i = -1, \dots, m$ such that

$$G = F^{-1} \subseteq F^0 \subseteq F^1 \subseteq \dots \subseteq F^{m-1} \subseteq F^m = 0$$

Then we define the **associated graded group of G with respect to the Filtration $\{F_i\}$** to be

$$\text{gr}(G) := \bigoplus_{i=0}^m F^{i-1} / F^i.$$

From $\text{gr}(G)$ we can rebuild G by rebuilding $\{F^i\}$ up to extension problems in the following way:

$$\begin{array}{ll}
F^{m-1} & = F^{m-1}/F^m = \text{gr}(G)_m \\
F^{m-2} & \text{ is an extension } 0 \longrightarrow F^{m-1} \longrightarrow F^{m-2} \longrightarrow F^{m-2}/F^{m-1} = \text{gr}(G)_{m-1} \longrightarrow 0 \\
F^{m-1} & \text{ is an extension } 0 \longrightarrow F^{m-2} \longrightarrow F^{m-3} \longrightarrow F^{m-3}/F^{m-2} = \text{gr}(G)_{m-2} \longrightarrow 0 \\
\vdots & \dots \\
F^{-1} & \text{ is an extension } 0 \longrightarrow F^0 \longrightarrow F^{-1} \longrightarrow F^{-1}/F^0 = \text{gr}(G)_0 \longrightarrow 0
\end{array}$$

Corollary 2.0.14. Define the groups

$$F^l K^n(M) := \ker(K^n(M) \rightarrow K^n(N_l)) .$$

Those give an decending filtration of the n -th K-group of M :

$$K^n(M) = F^{-1} K^n(M) \supseteq F^0 K^n(M) \supseteq \dots \supseteq F^{m-1} K^n(M) \supseteq F^m K^n(M) = 0 .$$

Then by the corollary above we can conculde that

$$E_r^{p,q} = \frac{F^{p-1} K^{p+q}(M)}{F^p K^{p+q}(M)} .$$

Hence, the diagonals on the infinity page are the associated graded groups to $K^{p+q}(M)$ with respect to the filtration above.

Corollary 2.0.15 (The first page). In our setting we get the first page to be defined as the quotient

$$E^{p,q} := K^{p+q}(N_p, N_{p-1})$$

Now since (N_p, N_{p-1}) is a regular index pair for $\text{Crit}_p(f)$ we have a flow induced homotopy equivalentce:

$$N_p/N_{p-1} \simeq \left(\bigcup_{w \in \text{Crit}_p(f)} D_\varepsilon^u(w) \right) / \left(\bigcup_{w \in \text{Crit}_p(f)} \partial D_\varepsilon^u(w) \right) \cong \bigvee_{w \in \text{Crit}_p(f)} S^p$$

Here, we use the definitions from above

$$D_\varepsilon^u(w) := \varphi_w^{-1}(\{v \in T_w^u M \mid \|v\| \leq \varepsilon\}),$$

and the last isomorphism is induced by orientations in the unstable manifolds. We can continue our inspection:

$$E_1^{p,q} \cong \tilde{K}^{p+q}(N_p/N_{p-1}) \tag{7}$$

$$\cong \bigoplus_{w \in \text{Crit}_p(f)} \tilde{K}^q(pt) = C^k(M, f, g, \mathfrak{o}; \tilde{K}^q(pt)) \tag{8}$$

$$(9)$$

Notice that this is trivial for all odd q and naturally isomorphic for all even q . In sum we have the chain of horizontal isomorphisms:

$$\begin{array}{ccccccc}
 E_1^{p,q} & \xrightarrow{\alpha} & \tilde{K}^{p+q}(N_p/N_{p-1}) & \xrightarrow{\beta} & \oplus_{w \in \text{Crit}_p(f)} \tilde{K}^q(pt) & \longleftarrow & C^p(M, \mathfrak{A}, \tilde{K}^q(pt)) \\
 \downarrow d_1^{p,q} & & \downarrow & & \downarrow & & \downarrow \\
 E_1^{p+1,q} & \xrightarrow{\alpha'} & \tilde{K}^{p+1+q}(N_{p+1}/N_p) & \xrightarrow{\beta'} & \oplus_{w \in \text{Crit}_{p+1}(f)} \tilde{K}^q(pt) & \longleftarrow & C^{p+1}(M, \mathfrak{A}, \tilde{K}^q(pt))
 \end{array}$$

Diagram 10

By naturality of the triple boundary operator, that the second vertical map is the boundary operator. Now on the next rug we can induce two maps. From the left to make the diagram commute and from the right. Do they agree? Or asked differently, is the map induced from the d_1 map the boundary operator of morse homology? We already answered this in theorem ?? . This theorem lets us define a new spectral sequence as follows:

Definition 2.0.16. Let $(D, E, \alpha, \beta, \gamma)$ be the exact couple from definition 2.0.1. Furthermore, let $\Psi := \oplus_{p,q} \Psi^{p,q}$ be the isomorphism from $E = \oplus_{p,q} E^{p,q} \rightarrow \oplus_{p,q} C^p(M, \mathfrak{A}, \tilde{K}^q(pt))$ that is defined in equation 7. Furthermore, let $\beta : K^{2s}(pt) \rightarrow \mathbb{Z}$ be the isomorphisms induced from the bott generator. Then we define

$$\Lambda : E \rightarrow \tilde{E} := \bigoplus_{p,q} C(M, \mathfrak{A}, \mathbb{Z}) \otimes_{\mathbb{Z}} \begin{cases} \mathbb{Z} & \text{if } q \text{ is even,} \\ 0 & \text{if } q \text{ is odd.} \end{cases}$$

Then define $\tilde{D} := D$ and the new exact couple $(\tilde{D}, \tilde{E}, \tilde{\alpha}, \tilde{\beta}, \tilde{\gamma})$ as follows:

$$\begin{aligned}
 \tilde{\alpha} &:= \alpha & : \tilde{D} &\rightarrow \tilde{D} \\
 \tilde{\beta} &:= \Lambda \circ \beta & : \tilde{D} &\rightarrow \tilde{E} \\
 \tilde{\gamma} &:= \gamma \circ \Lambda^{-1} & : \tilde{E} &\rightarrow \tilde{D}
 \end{aligned}$$

We get a morphism of exact couples from $(D, E, \alpha, \beta, \gamma)$ to $(\tilde{D}, \tilde{E}, \tilde{\alpha}, \tilde{\beta}, \tilde{\gamma})$ that is given as (Λ, id_D) . This new exact couple gives rise to a spectral sequence $(\tilde{E}_r, \tilde{d}_r)_r$, where the first page is the Morse-chain group in all even q . Hence, the second page is the Morse homology.

Corollary 2.0.17 (Periodicity in q). The exact couple $(\tilde{D}, \tilde{E}, \tilde{\alpha}, \tilde{\beta}, \tilde{\gamma})$ has periodicity in q meaning that we have the isomorphism of exact couples given by the identity functions

$$f^{p,q} : \tilde{E}^{p,q} \rightarrow \tilde{E}^{p,q+2}$$

Here the notation might seem a bit strange. However, the summands $\tilde{E}^{p,q}$ and $\tilde{E}^{p,q+2}$ are actually the same. They only differ by there inclision map into $\tilde{E}$. Now the pair $(f : \oplus_{p,q} \tilde{E} \rightarrow \tilde{E}, g := \text{id } \tilde{D} \rightarrow \tilde{D})$ defines an isomorphism of exact couples. The compatibility with morphisms is clear. Now since building up the spectral sequence is functorial we

know that for all p, q we have canonical (in the sence that they traslate $\widetilde{d}_r^{p,q}$ to $\widetilde{d}_r^{p,q+2}$) isomorphisms

$$\widetilde{E}_r^{p,q} \cong \widetilde{E}_r^{p,q+2}.$$

Furthermore, $\widetilde{E}_r^{p,q} = 0$ for all odd q .

Definition 2.0.18. For $r > 0$ we define $EM_r^p := \widetilde{E}_r^{p,0}$ and

$$dM_r : EM_r^p \rightarrow EM_r^{p+r}; x \mapsto \begin{cases} 0 & \text{if } r \text{ is even} \\ \widetilde{d}_r^{p,0}(x) \in \widetilde{E}_r^{p,-r+1} = EM_r^p & \text{else.} \end{cases}$$

The last equality comes from the corollary above. Furthermore, we have a morphism of spectral sequences $(\widetilde{E}_r, \widetilde{d}_r) \rightarrow (EM_r, dM_r)$ by the maps

$$f_r^{p,q} : \widetilde{E}_r^{p,q} \rightarrow EM_r^p; x \mapsto \begin{cases} 0 & \text{if } q \text{ is odd} \\ x \in EM_r = \widetilde{E}_r^{p,0} \cong \widetilde{E}_r^{p,q} & \text{if } q \text{ is even.} \end{cases}$$

Theorem 2.0.19 (Summary of the Spectral Sequence). *For each Morse datum $(M, f, g, \mathfrak{A})$ together with a filtration $N_{-1} \subseteq N_0 \subseteq \dots \subseteq N_{m-1} \subseteq N_m = M$ from (6) there exists a Spectralsequence (EM_r^p, dM_r^p) where each page is graded in p such that*

- *The first page is the morse complex, hence the second page is the Morse homology*
- *The spectral sequence becomes stationary and the infinity page is the associated graded group of the K -groups meaning:*

$$\bigoplus_{p \in 2\mathbb{N}} E_\infty^p \text{ is a associated graded group to } K^0(M)$$

$$\bigoplus_{p \in 2\mathbb{N}+1} E_\infty^p \text{ is a associated graded group to } K^1(M).$$

Remark 2.0.20 (Recover from Associated Graded Groups). Let G be a group and

3 Higher Differentials

4 Examples

We begin with a minimal example that has non trivial higher differential:

Example 4.0.1 (The Sphere). If we take the three-sphere $S^3 \subseteq \mathbb{R}^4$ and denote the north pole with N and the south pole with S we get the Morse chain complex as

$$0 \longrightarrow \langle S \rangle_{\mathbb{Z}} \longrightarrow 0 \longrightarrow 0 \longrightarrow \langle N \rangle_{\mathbb{Z}} \longrightarrow 0$$

Then for the second page we get the Morse homology together with the 0 maps and on the third page we have:

$$\begin{array}{ccc} \langle S \rangle & \begin{array}{cc} 0 & 0 \end{array} & \langle N \rangle \\ & \searrow \quad \quad \quad \nearrow & \\ & d_3^0 & \end{array}$$

Now we want to check what $d_3^0(S)$ is. For this we want to use its definition and hence we need to give the Morse filtration. For this we can use:

$$N_{-1} := \emptyset, \quad N_{0,1,2} := H_- \text{ the southern hemisphere}, \quad N_3 = S^3.$$

First we translate E_r into the language where the definition was formulated. We know that EM_3^p is naturally isomorphic to

$$E_3^{p,0} = \frac{\gamma^{-1}(\alpha^2 D^{p,-2})}{\beta(\ker(\alpha^2|_{D^{p+3-1,-2}}))}$$

Now for $r = 3$ and $p = 0$

$$\alpha^2 D^{2,-2} = \text{im}(K^0(N_2) \rightarrow K^0(N_0)) = \text{im}(K^0(H_-) \rightarrow K^0(H_-)) \cong \mathbb{Z}$$

Hence $E_3^{0,0} = \gamma^{-1}(\alpha^2 D^{2,-2})^{0,0} \cap E_0^{0,0} = K^0(N_0, \emptyset) \cong \mathbb{Z}$. And similar we conclude that $E_3^{3,0} = K^3(N_3, N_2) \cong \mathbb{Z}$. Now we analyze the map

$$d_3^{3,0} : K^3(N_3, N_2) \rightarrow K^0(N_0, \emptyset)$$

So far we were able to ignore all orientation, (which we formally need for the morse boundary operator). So let's fix the orientations of the unstable part by saying the unstable part of the north pole (i.e. the whole S^3) is the canonical orientation of $S^3 \subset \mathbb{R}^4$, whilst the unstable part of the south pole (i.e. the south pole) has the orientation -1 .

Now $d_r = \beta \circ \alpha^2 \circ \gamma$ and $\gamma : K^3(N_3, N_2) \rightarrow K^0(N_3)$ maps our generator induced by N to the standard generator of $K^3(S^3)$ induced by the chosen orientation, due to N_2 being a contractible subset. Now $\alpha^2 : K^3(N_3) \rightarrow K^3(N_1)$ is the zero map, since its range is 0 due to N_1 being homotopically equivalent to the point and 3 being odd. Finally $\beta : K^3(N_1) \rightarrow K^4(N_0, \emptyset)$ is the connecting homomorphism that is trivial here. Hence $d_3^{3,0}$ is the zero map. And thereby EM_4 which equals E_∞ is

$$\langle S \rangle \quad 0 \quad 0 \quad \langle N \rangle$$

Now how to we get information about our beloved K groups back? For this we remind ourself about the filtration

$$K^0(S^3) = F^{-1}K^0(S^3) \supseteq F^0K^0(S^3) \supseteq F^1K^0(S^3) \supseteq F^2K^0(S^3) \supseteq F^3K^0(S^3) = 0$$

Now using this filtration we have the short exact sequence

$$0 \longrightarrow F^0K^0(S^3) \longrightarrow K^0(S^3) \longrightarrow E_4^{0,0} = \frac{F^{-1}K^0(S^3)}{F^0K^0(S^3)} \longrightarrow 0$$

Hence, $K^0(S^3)$ is an extension of $F^0K^0(S^3)$ and $E_4^{0,0} \cong \mathbb{Z}$. Furthermore, we have the exact sequence

$$0 \longrightarrow F^1K^0(S^3) \longrightarrow F^0K^0(S^3) \longrightarrow E_4^{1,-1} = \frac{F^0K^0(S^3)}{F^1K^0(S^3)} = 0 \longrightarrow 0$$

Hence, $F^0K^0(S^3) \cong F^1K^0(S^3)$ and by the same argument we can continue this chain

$$F^0K^0(S^3) \cong F^1K^0(S^3) \cong F^2K^0(S^3) \cong F^3K^0(S^3) = 0$$

Hence, the first short exact sequence gives is an isomorphism between $K^0(S^3) \cong \mathbb{Z}$.

Now we want to calculate the K group of a non trivial example:

Example 4.0.2 (The Real Projective Space). Here, we define $\mathbb{RP}^n$ to be $\mathbb{R}^{n+1} \setminus 0$ modulo scalar multiples. Hence it is the space of lines in $\mathbb{R}^{n+1}$. Alternatively we can view it as the n -sphere where the operation from $\{\text{id}, -\text{id}\}$ is quotient out. From this description we get a metric, since $S^n \supseteq \mathbb{R}^{n+1}$ inherits the subspace metric in which all operations from $\{\text{id}, -\text{id}\}$ are isometries. Hence, this metric projects down to $\mathbb{RP}^n$. Now let

$$A = \text{diag}(\lambda_0, \lambda_1, \dots, \lambda_n)$$

be a symmetric matrix with real $\lambda_i \in \mathbb{R}$. Then we define the map

$$\Lambda : \mathbb{R}^{n+1} \setminus \{0\} \rightarrow \mathbb{R}; \Lambda(x) := \frac{\langle x, Ax \rangle}{\|x\|^2}.$$

Notice that this function descends to a well defined function $\tilde{\Lambda} : \mathbb{RP}^n \rightarrow \mathbb{R}$. On the whole $\mathbb{R}^{n+1}$ we can define the function

$$f : \mathbb{R}^{n+1} \rightarrow \mathbb{R}; f(x) := \frac{1}{2} \langle x, Ax \rangle = \frac{1}{2} \sum_{i=0}^n \lambda_i x_i^2.$$

Now assume that $\varphi_t(x)$ is the flow associated to $-\text{grad}(f)$, meaning

$$\frac{d\varphi_t(x)}{dt} = -\text{grad}_{\varphi_t(x)}(f) = -A\varphi_t(x),$$

since

$$\frac{\partial f}{\partial x_i}(x) = \lambda_i x_i.$$

Now we have the lemma:

Lemma 4.0.3. *The flow lines of the negative gradient flow with respect to $\frac{1}{2}\tilde{\Lambda}$ on $\mathbb{RP}^n$ are the images under the quotient map*

$$\pi : \mathbb{R}^{n+1} \setminus \{0\} \rightarrow \mathbb{RP}^n$$

of the flow lines of the linear system

$$\frac{d\Phi_t(x)}{dt} = -A\Phi_t(x)$$

Proof. Let x be a point on the flow line $\Phi_t(x)$. Then $-\text{grad}(f)(x) = -Ax$. Now we split this into a part that is orthogonal to y and one that is parallel:

$$-\text{grad}(f)(x) = \underbrace{-\Lambda(x)x}_{\text{parallel to } x} + \underbrace{-Ax + \Lambda(x)x}_{\text{orthogonal to } x}.$$

The claim of orthogonality is just a calculation of the inner product $\langle -Ax + \Lambda(x)x, x \rangle = 0$ $\square$

Now we radially project this onto the sphere via

$$\pi_r : \mathbb{R}^{n+1} \setminus \{0\} \rightarrow S^n; x \mapsto \frac{x}{\|x\|}$$

the differential of this projection kills the parallel part and hence reads:

$$d\pi_r|_x(-\text{grad}_x(f)) = \frac{-Ax + \Lambda(x)x}{\|x\|}$$

If we restrict $f|_{S^n}$ we get that the gradient $-\text{grad}_y(f|_{S^n}) = -Ay + \Lambda(y)y$. To see this, let $v \in T_y S^n \subseteq T_y \mathbb{R}^{n+1}$ and remember that the inner product on S^n is induced from the euclidean one. Then

$$\begin{aligned} \langle \text{grad}_y(f|_{S^n}), y \rangle_{S^n} &= d f|_{S^n}(y) = df(y) = \langle \text{grad}_y(f), y \rangle_{\mathbb{R}^{n+1}} \\ &= \langle \text{proj}_{T_y S^n}(\text{grad}_y(f)), y \rangle_{\mathbb{R}^{n+1}} = \langle Ay - \Lambda(y)y, y \rangle_{\mathbb{R}^{n+1}} \end{aligned}$$

Here $\text{proj}_{T_y S^n}$ denotes the projection onto the subspace $T_y S^n \subseteq T_y \mathbb{R}^{n+1}$. This however tells us that the image $\widetilde{\Phi_t(x)}$ of $\Phi_t(x)$ under the radial projection solves the equation

$$\frac{d\widetilde{\Phi_t(y)}}{dt} = -\text{grad}_y(f|_{S^n})(\widetilde{\Phi_t(y)}) = -\frac{1}{2}\text{grad}_y(\Lambda|_{S^n})(\widetilde{\Phi_t(y)}) \quad \text{since } f = \frac{1}{2}\Lambda$$

Hence descending to the quotient $\mathbb{RP}^n$ yields that $\left[\widetilde{\Phi_{2t}(y)}\right]$ is a solution of the flow associated to the negative gradient of $\tilde{\Lambda}$ in $\mathbb{RP}^n$. Now the other way round should be trivial: Take a solution $\psi_t(x)$ of $\tilde{\Lambda}$ in $\mathbb{RP}^n$. Then for fixed t, x let U be a neighbourhood

of $\psi_t(x)$. This U can be lifted to S^n and the solution there solves the lifted version of the flow induced from $\tilde{\Lambda}$. And from here we can argue that the solution is part of a solution of the linear differential equation

$$\frac{d\varphi_t(x)}{dt} = -\text{grad}_{\varphi_t(x)}(f) = -Ax.$$

This concludes the proof. Hence critical points in $\mathbb{RP}^n$ are eigenvalues of A . Hence, for the function to be Morse, we need the critical points to be isolated, meaning that all eigenspaces have dimension one. So we assume that the diagonal entries in A are all different. For simplicity we assume that $\lambda_0 < \lambda_1 < \dots < \lambda_n$. Since A is diagonal we know that $[e_i]$ are the critical points.

Lemma 4.0.4. *Each critical point $[e_i]$ is non degenerate, of index i and its unstable manifold is*

$$W([e_i] \rightarrow) =: U_i = \{[x_0 : x_1 : \dots : x_i : 0 : \dots : 0] \in \mathbb{RP}^n | x_i \neq 0\},$$

whilst the stable manifold is

$$W(\rightarrow [e_i]) =: S_i = \{[0 : \dots : 0 : x_i : x_{i+1} : \dots : x_n] \in \mathbb{RP}^n | x_i \neq 0\}$$

Proof. Let B^n be the open unit ball in $\mathbb{R}^n$ and $V_i := \{[x, \dots, x_n] \in \mathbb{RP}^n | x_i \neq 0\}$. Then we define the parametrization

$$\begin{aligned} \varphi_i : B^n &\rightarrow V_i \\ (y_1, \dots, y_n) &\mapsto [y_1, \dots, y_i, \sqrt{1 - \|y\|_2^2}, y_{i+1}, \dots, y_n]. \end{aligned}$$

Now the image gives representatives of norm one and hence in this chart we calculate:

$$\begin{aligned} \tilde{\Lambda} \circ \varphi_i(y_1, \dots, y_n) &= \sum_{j=1}^i y_j^2 \lambda_{j-1} + \lambda_i (1 - \|y\|_2^2) + \sum_{j=i+1}^n y_j^2 \lambda_j \\ &= \sum_{j=1}^i y_j^2 \lambda_{j-1} + \lambda_i (1 - \sum_{j=1}^n y_j^2) + \sum_{j=i+1}^n y_j^2 \lambda_j \\ &= \sum_{j=1}^i y_j^2 (\lambda_{j-1} - \lambda_i) + \lambda_i + \sum_{j=i+1}^n y_j^2 (\lambda_j - \lambda_i) \end{aligned}$$

With this local description we can calculate the hessian in $[e_i] = \varphi_i(0)$:

$$\left. \frac{\partial^2 \tilde{\Lambda} \circ \varphi_i}{\partial y_k \partial y_l} \right|_0 = \begin{cases} 0 & \text{if } k \neq l \\ 2(\lambda_{k-1} - \lambda_i) < 0 & \text{if } k = l \leq i \\ 2(\lambda_k - \lambda_i) > 0 & \text{if } k = l > i \end{cases}$$

Hence, $[e_j]$ is a non degenerate critical point of index i . Now concerning the unstable manifolds, we pick a point $x = x_k e_k + \dots x_l e_l \in \mathbb{R}^{n+1}$ with x_k and x_l not equal to zero. This points trajectory in $\mathbb{R}^{n+1}$ under $-\text{grad}(f)$ is

$$\varphi_t(x) = x_k e^{-\lambda_k t} e_k + \dots x_l e^{-\lambda_l t} e_l.$$

Hence for t approaching negative infinity, the term $x_l e^{-\lambda_l t} e_l$ is leading, whilst for t approaching infinity, the summand $x_k e^{-\lambda_k t} e_k$ is leading. Since projectively we don't care about absolute values but rather relations having a leading coordinate meaning that relatively all other coordinates approach zero. Hence:

$$\lim_{t \rightarrow -\infty} ([\varphi_t(x)]) = [e_l] \quad \text{and} \quad \lim_{t \rightarrow +\infty} ([\varphi_t(x)]) = [e_k].$$

□

Definition 4.0.5 (Orientations). For orienting the unstable manifolds we make use of the projection $\mathbb{R}^{n+1} \supseteq S^n \rightarrow \mathbb{RP}^n$. Hence, we can lift $T_{[e_i]}\mathbb{RP}^n$ to $T_{\pm e_i}S^n$ and include that into $T_{\pm e_i}\mathbb{R}^{n+1}$, where we have the maps $Q_{\pm e_i} : \mathbb{R}^{n+1} \rightarrow T_{\pm e_i}\mathbb{R}^{n+1}$. In this way we get two canonical lifts of the tangent space:

$$\begin{array}{ccc} T_{[e_i]}\mathbb{RP}^n & \xrightarrow{d\pi_{e_i}^{-1}} & T_{e_i}S^n \\ \downarrow d\pi_{-e_i}^{-1} & & \downarrow Q_{e_i}^{-1} \\ T_{-e_i}S^n & \xrightarrow{Q_{-e_i}} & \langle (e_j)_{j \neq i} \rangle \end{array}$$

In general, the diagram is not commutative. To see this notice how $\pi \circ -\text{id} = \pi$. Hence $d\pi|_{-e_i} \circ d - \text{id}|_{e_i} = d\pi|_{e_i}$ and with that:

$$\begin{aligned} Q_{-e_i}^{-1} \circ d\pi|_{-e_i}^{-1} \circ d\pi|_{e_i} \circ Q_{e_i} &= Q_{-e_i}^{-1} \circ d\pi|_{-e_i}^{-1} \circ d\pi|_{-e_i} \circ d(-\text{id})|_{e_i} \circ Q_{e_i} \\ &= Q_{-e_i}^{-1} \circ d\pi|_{-e_i}^{-1} \circ d\pi|_{-e_i} \circ Q_{-e_i} \circ -\text{id} \\ &= Q_{-e_i}^{-1} \circ Q_{-e_i} - \text{id} = -\text{id}. \end{aligned}$$

Hence, we can define a canonical orientation on $T_{[e_i]}\mathbb{RP}^n$ that is given by $(e_j)_{j \neq i}$ or equivalent $(-e_j)_{j \neq i}$ depending on the chosen lift. Now in the same way we can induce orientations to the unstable manifolds U_i by two equivalent ways:

$$\begin{array}{ccc} T_{[e_i]}U_i & \hookrightarrow & T_{[e_i]}\mathbb{RP}^n \\ \uparrow & & \uparrow \\ \langle (e_0, \dots, e_{i-1}) \rangle & \hookrightarrow & T_{e_i}S^n \end{array} \quad \text{or equivalently} \quad \begin{array}{ccc} T_{[e_i]}U_i & \hookrightarrow & T_{[e_i]}\mathbb{RP}^n \\ \uparrow & & \uparrow \\ \langle -e_0, \dots, -e_{i-1} \rangle & \hookrightarrow & T_{-e_i}S^n \end{array}$$

Now we know how the groups in the Morse chain complex are looking. The next step is the boundary operator:

Lemma 4.0.6. *The differential in $C^*(\mathbb{RP}^n, \Lambda^*)$ with the right orientations act as*

$$\partial_i([e_i]) = n([e_{i-1}], e_i)[e_{i+1}] = ((-1)^i - 1)[e_{i+1}]$$

Proof. First we notice that $W([e_i] \rightarrow [e_{i-1}])$ is a one dimensional projective space with two points missing. If we would look at the Morse homology with coefficients in $\mathbb{Z}_2$ we would be done here, since $\mathbb{RP}^1 \setminus \{p, q\}$ consists of two lines. Hence, over $\mathbb{Z}_2$ the boundary operator is trivial. However we want to use the orientations from above. Now let

$$x \in W([x_i] \rightarrow [e_{i-1}]) = \{[0 : \dots : 0 : x_{i-1} : x_i : 0 : \dots : 0] \mid x_{i-1} \neq 0 \text{ and } x_i \neq 0\}.$$

Then we can choose representatives, such that $x_{i-1}^2 + x_i^2 = 1$, and (x_{i-1}, x_i) is in the upper halve plane meaning that $x_i > 0$. Now take a $x \in W([e_i] \rightarrow [e_{i-1}])$. Then $T_x W([e_i] \rightarrow [e_{i-1}])$ has an oriented basis given as the pushforward of $v = x_i e_{i-1} - x_{i-1} e_i$ under the projection $T_x S^n \rightarrow T_x \mathbb{RP}^n$. Now we want to analyze how this relates to the orientation induced by the flow and the two unstable orientations: if both x_i and x_{i-1} are positive, then v points in the direction of the flow or said different, $-\text{grad}_x(f) = v$. Else, if x_{i-1} is negative, $-v$ points in the direction of the flow (i.e. $-\text{grad}_x(f) = -v$). The parallel transport of $T_{[e_{i-1}]} U_{i-1}$ to the normal space of $-\text{grad}_x(\tilde{\Lambda})$ induced by the negative flow maps the basis $(e_0, \dots, e_{i-2})$ of $T_{[e_{i-1}]} U_i$ to $(e_0, \dots, e_{i-2})$ in the orthogonal complement of $-\text{grad}_x(\tilde{\Lambda})$ in $T_x \mathbb{RP}^n$. So we need to compare in $T_x \mathbb{RP}^n$ the basis $(-\text{grad}_x(f), e_0, \dots, e_{i-2})$ to the basis $(e_0, \dots, e_{i-1})$ in $T_{[e_i]} U_i$ after transporting it to $T_x \mathbb{RP}^n$ along the flow. This latter transport maps the basis vectors as follows:

$$e_j \mapsto \begin{cases} x_i e_{i-1} - x_{i-1} e_i & \text{if } j = i-1 \\ e_j & \text{else.} \end{cases}$$

In other words, we need to ask when the two basis

$$((x_i e_{i-1} - x_{i-1} e_i), e_0, \dots, e_{i-2}) \quad \text{and} \quad (e_0, \dots, e_{i-2}, x_i e_{i-1} - x_{i-1} e_i),$$

if $x_{i-1} > 0$ are comparable. Then, the basis carry the same orientation, if and only if i is odd, since then it takes evenly many transpositions of coordinates.

If $x_{i-1} < 0$ we notice, that our representation of $[e_{i-1}]$ is given by $-e_{i-1}$ and hence the unstable tangent space has the basis $-e_0, \dots, -e_{i-2}$. Hence in x we get the oriented basis

$$(-\text{grad}_x(f), -e_0, \dots, -e_{i-2}) = (-v, -e_0, \dots, -e_{i-2}).$$

We compare this basis to basis that comes from transporting the basis $(e_0, \dots, e_{i-1})$ of $T_{[e_i]} U_{[e_i]}$ to x resulting in the basis $(e_0, \dots, e_{i-2}, v)$. Hence we compare the bases

$$(-v, -e_0, \dots, -e_{i-2}) \quad \text{and} \quad (e_0, \dots, e_{i-2}, v)$$

we manipulate the first and keep note of the sign of the alterations:

$$(-v, -e_0, \dots, -e_{i-2}) = (-1)^i (v, e_0, \dots, e_{i-2}) = (-1)^{i+i-1} (e_0, \dots, e_{i-2}, v).$$

Hence, the orientations never agree. Now if i is odd the two flow lines cancel out when counted with the sign and if i is even, we count minus two flow lines. Therefore the counting of flow lines noted by $n(p, q)$ gives:

$$n(e_{i+1}, e_i) = (-1)^i - 1,$$

and the Morse cohomology boundary operator equals:

$$\partial_i([e_i]) = n([e_{i-1}], e_i)[e_{i+1}] = ((-1)^i - 1)[e_{i+1}]$$

□

Now for even n , EM_1 of the form:

$$0 \quad \langle [e_0] \rangle \xrightarrow{0} \langle [e_1] \rangle \xrightarrow{-2} \langle [e_2] \rangle \xrightarrow{0} \dots \xrightarrow{-2} \langle [e_n] \rangle \xrightarrow{0} 0$$

and for odd n :

$$0 \quad \langle [e_0] \rangle \xrightarrow{0} \langle [e_1] \rangle \xrightarrow{-2} \langle [e_2] \rangle \xrightarrow{0} \dots \xrightarrow{0} \langle [e_n] \rangle \xrightarrow{0} 0$$

Hence we have the EM_2 pages as follows:

$$0 \quad \langle [e_0] \rangle \quad 0 \quad \frac{\langle [e_2] \rangle}{\langle -2[e_2] \rangle} \quad 0 \quad \dots \quad \frac{\langle [e_n] \rangle}{\langle -2[e_n] \rangle} \quad 0$$

if n is even, and in the odd case:

$$0 \quad \langle [e_0] \rangle \quad 0 \quad \frac{\langle [e_2] \rangle}{\langle -2[e_2] \rangle} \quad 012 \quad \dots \quad \langle [e_n] \rangle \quad 0$$

When it comes to higher differentials, we notice the following: Only for odd r the differentials are possible nontrivial. Hence for $\mathbb{RP}^2$ we are done and we can get the complex k-groups: For $K^0(\mathbb{RP}^2)$ we get an associated graded group that is isomorphic to $\mathbb{Z} \oplus \mathbb{Z}_2$. Hence we need to calculate the extension

$$0 \longrightarrow \mathbb{Z} \longrightarrow K^0(\mathbb{RP}^2) \longrightarrow \mathbb{Z}_2 \longrightarrow 0$$

giving us $K^0(\mathbb{RP}^2) \cong \mathbb{Z} \oplus \mathbb{Z}_2$. And similar $K^1(\mathbb{RP}^2) = 0$. To understand the higher differentials we need to get specific about the used filtration. We define the filtration inductively as follows: For any i define $c_i \in (\lambda_i, \lambda_{i+1})$ and with this we define the filtration

$$N_{-1} := \emptyset, \quad N_i = \tilde{\Lambda}(-\infty, c_i) \text{ the sublevel sets, } \quad N_n := \mathbb{RP}^n.$$

With this filtration it is somewhat easy to see that (N_i, N_{i-1}) gives a regular index pair of $[e_i]$. One just needs to go through the definitions: First it is clear, that the only critical point inside of the closure of $N_i \setminus N_{i-1}$ is $[e_i]$ and it is in the interior. Secondly, the positive line invariance is clear, since $\tilde{\Lambda}$ decreases along flow lines. And N_{i-1} is an exit set also by definition. Finally the cofibration property of the pair N_i, N_{i-1} is to be shown. For this we notice, that c_i is a regular value of R . Hence there exists an $\varepsilon > 0$, such that $f^{-1}([a, a + \varepsilon))$ contains no critical point and hence this defines an open collar of N_i inside N_{i+1} . This open collar can be shown to be an NDR pair, which in turn is a cofibration. Those are standard arguments seen in [Whi78] Chapter 5.1.

Lemma 4.0.7. *In the filtration, each N_i is homotopically equivalent to $\mathbb{RP}^i$.*

Proof. Notice how for all i the sublevelset $\tilde{\Lambda}^{-1}(-\infty, \lambda_i]$ is an embedded projective space $\mathbb{RP}^i$. To show that N_i is homotopically equivalent we start by noticing that $N_i \simeq \tilde{\Lambda}^{-1}(-\infty, \lambda_{i-1} + \varepsilon)$ for all $\varepsilon > 0$ small enough. Now the next step is to get rid of epsilon. Let L_ε denote $\tilde{\Lambda}^{-1}(-\infty, \lambda_{i-1} + \varepsilon)$ and hence we want to show that $L_\varepsilon \simeq L_0$. For this we define

$$H : L_\varepsilon \times [0, 1] \rightarrow L_\varepsilon \quad ([x], t) \text{ with } \|x\| = 1 \mapsto \left[\sum_{j=1}^i x_j e_j + (1-t) \sum_{j>i} x_j e_j \right]$$

First we need to check if this map is well defined in two senses. Is the definition invariant under the chosen representative and do we map into L_ε . For the independence of the chosen representative we notice that the image of x and $-x$ differ by sign and hence define the same equivalence class. Also applying $\tilde{\Lambda}$ to the image we notice that the image is less than $\lambda_i + \varepsilon$. this is true, since $\tilde{\Lambda}$ is a weighted sum of the eigenvalues and the addition of $(1-t)$ decreases the weights on the bigger eigenvalues reducing the overall weighted sum.

Now obviously H_1 maps to $\tilde{\Lambda}^{-1}(-\infty, \lambda_i]$. □

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Now that we have our filtration we can start looking at higher differentials. If n is even, then every other group in EM_2 is zero hence all higher differentials vanish as only odd d_r are candidates to be non trivial. So let n be odd. Then only differentials landing in the last group EM_n can possibly be non trivial. However, for all $r < n$ the differential that lands in $EM_r^m \cong \mathbb{Z}$ starts in $EM_r^{m-r} \cong \mathbb{Z}_2$. Hence they are all trivial as well. Now if $r = m$ we have the differential

$$d_m^0 : EM_m^0 \rightarrow EM_m^m$$

which is possibly non-zero. However this differential is given as follows:

$$\begin{array}{ccccc} & & & & K^0(N_m) \\ & & & & \downarrow i^* \\ K^0(N_0, \emptyset) & \xrightarrow{\gamma} & K^0(N_0) & \xrightarrow{\widetilde{\alpha^{-m}}} & K^0(N_{m-1}) \\ & \searrow d_m^0 & & & \downarrow \delta \\ & & & & K^1(N_m, N_{m-1}) \end{array}$$

Here α^{-m} is any left inverse of α^m or taking the preimage (the well definedness or rather independence of the chosen inverse was shown in the definition of the derived exact couple). The vertical sequence is exact and since $N_{m-1} = N_m \setminus D$ where D is contractible, we have that i^* is surjective and hence δ is trivial and thereby $dM_m^0 = 0$. Then for this we start with $n = 3$ and then go up. In the case of $n = 3$ we are trivial, since d_3^0 is given by the composition

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$$\begin{array}{ccccc}
& & & K^0(N_3) & \\
& & & \downarrow i^* & \\
K^0(N_0, \emptyset) & \xrightarrow{\gamma} & K^0(N_0) & \xrightarrow{\widetilde{\alpha^{-2}}} & K^0(N_2) \\
& \searrow d_3 & & \downarrow \delta & \\
& & & K^1(N_3, N_2) &
\end{array}$$

Here α^{-1} is any left inverse of α^2 or taking the preimage (the well definedness or rather independence of the chosen inverse was shown in the definition of the derived exact couple). The vertical sequence is exact and since $N_2 = N_3 \setminus D$ where D is contractible. A contraction is given by flowing along the positive gradient of $\tilde{\Lambda}$. Hence, we have that i^* is surjective and thereby δ is trivial resulting in $dM_m^0 = 0$. For all real projective spaces the spectral sequence becomes stable on the second page and hence we can calculate the complex K-groups:

Example 4.0.8. Make a produkt morse funktion on $\mathbb{RP}^2 \times \mathbb{RP}^4$!

References

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